

INTERNSHIPS AND WORK EXPERIENCE

Locust Occurrence Modelling and Prediction

Summer 2021

Data Science Associate | Data Science Intern

Dtime

- Worked on an **end-to-end** machine learning project to model and forecast **locust outbreaks** in East-African region.
- Implemented machine learning algorithms to **extract and engineer new features** and capture the **spatial and temporal** characteristics of the problem, and tackle the heavily skewed nature of the problem, (*as high as 1:5k*).
- Implemented **Bi-directional ConvLSTM** based **U-Net** model for pixel level classification on **67k grid-IDs** of $10km^2$ each, projected over spatial images corresponding to **450+ timestamps**, with respective segmentation masks.
- Used a combination of algorithms like **SMOTEENN** and **SMOTETomek**, with **Balanced Ensemble Classifiers** to reduce the effect of class asymmetry and predict a locust occurrence, **minimizing false negatives** successfully.

Japanese Document OCR Pipeline

Summer 2022

Data Science Intern

Daikin

- Developed an **end-to-end OCR** system capable of text detection, recognition and information extraction.
- Enhanced the limited Japanese dataset using **document-centric** image processing and data augmentation techniques.
- Trained model based on **MobileNetV3 backbone** using **DB algorithm** for text detection to get **F-score** of **0.9**.
- Implemented and finetuned a **CRNN model** for real-time recognition and achieve an **Edit-Distance-Score** of **0.91**.
- Analysed methods for information extraction including Graph Convolution Networks and Spatially-aware transformers.

ACADEMIC PROJECTS

Spoof-Resistant Face Recognition

Summer 2020

Institute Technical Summer Project

WnCC, IIT Bombay

- Developed a system to differentiate between live and digitally created faces from input image and identify them.
- Implemented **OpenCV's Caffe** based Single Shot Detector (SSD) framework, to get the region of interest.
- Built an **artificial neural network** incorporating several **convolutional** layers in combination with others and trained it on a **self-made** live and fake images dataset, for **liveness detection feature**.
- Extracted **128-D facial encoding vectors** from **dlib's 5-point facial landmark model** to train a support vector machine and work with the liveness detection model for **facial recognition capability**.

Indic Handwriting Recognition

Spring 2022

Prof. Ganesh Ramakrishnan — (RnD project)

IIT Bombay

- Worked on an RnD project trying to model and solve the problem of **Handwriting Recognition** for **Indic languages**.
- Implemented a model using convolutional and **Bidirectional-LSTM** layers and **CTCloss** for English language.
- Studied and touched some topics like subset selection, knowledge distillation through various research papers.

Lossless High-Entropy Compression Algorithm

Summer 2020

Seasons of Code

WnCC, IIT Bombay

- Achieved the task to convert numeric sequences of a fixed length to compressed versions minimizing the complexity of their **fourier-transform**, achieving **high entropy data compression**.
- Implemented a system of **seq2seq recurrent neural network** using *Long Short Term Memory (LSTM)* units in combination with other layers to output compressed form and make transmission and storage more viable.
- Generated a mapping using the compressed signal to **recreate original signal** for simpler signals accurately.

Quantum Information and Computing

Summer 2020

Summer of Science

MnP Club, IIT Bombay

- Completed a reading project on mathematical aspects of **Quantum Information and Computing**.
- Studied differences in computational methods in classical and quantum computers.
- Learned about some basic **quantum algorithms** and the computational ease offered by them over classical counterparts for some tasks known to be computationally complex for classical computers.

Automatic Rule Induction - SPEAR

Prof. Ganesh Ramakrishnan — (Ongoing B.Tech. project)

Spring 2022

IIT Bombay

- Working to Extend the **SPEAR** library to include automatic rule induction approaches.
- Researching and evaluating ways to help integrate existing methods within the SPEAR framework to widen the scope of the library and ensure robust evaluation of weak supervision approaches.

Red Plag : Plagiarism Checker

Prof. Amitabha Sanyal — (Course project)

Autumn 2020

IIT Bombay

- Developed a **plagiarism checker** to measure pairwise similarity between a collection of text files.
- Adopted an algorithm that involved using **separate tokenizers** for languages **C++**, **Java**, **Python** followed by **winnowing of vectors** from **hashed k-grams** to compute similarity percentage.
- Created a web **front-end** using **Angular** framework to visualize results in a graphical manner, and **Django** based web-framework for the **backend** server to maintain the database, and to link and store query results.
- Added **authentication** using *JSON web-tokens* (**JWT**), to restrict access to authorised individuals only.
- Added functionality for use with given boilerplate code, while comparing multiple files, for accuracy.

C-like language Compiler

Prof. Uday Khedker — (Course Project)

Spring 2021

IIT Bombay

- Built a compiler that could return **Assembly line** code from a C-like program combining Lex and Yacc.
- Coded a scanner and parser that could produce the **Three Address Code** using the **Abstract Syntax Tree**.
- Added the functionality to produce Register Transfer Language using the previously produced outputs of the program.

SCHOLASTIC ACHIEVEMENTS

- Secured **99.84** percentile in **JEE Mains** out of over **1.1 million** candidates (2019)
- Achieved **98.3** percentile in **JEE-Advanced** out of more than **0.2 million** candidates (2019)

POSITIONS OF RESPONSIBILITY

Institute Student Mentorship Program (ISMP) Mentor

2022 - 2023

- Selected to a team of 144-members out of 388 applicants via a rigorous interview process and peer reviews.
- Responsible to mentor a group of first year students and help & guide them through various difficulties.

Department Academic Mentorship Program (DAMP) Mentor

2021 - 2022 - Present

Department of CSE, IIT Bombay

- Appointed as **Department Academic Mentor** for two years running. Selected after a process of rigorous interviews and extensive peer reviews. Responsible to guide students in the Department through academic and personal difficulties.

Institute Technical Summer Project (ITSP) Mentor

Summer 2021

- Guided a team of 5 members for a machine learning project involving a facial and speech recognition system.

TECHNICAL SKILLS

Programming:	C++, C, Python, Java, Matlab, VHDL, HTML, CSS, SQLite, Bash, Socket, z3
Development:	Android Studio, Angular, Django, Git, L ^A T _E X, Arduino, CSS, JavaScript
Data Science & ML:	Pandas, Matplotlib, Sklearn, Imblearn, Pytorch, Tensorflow, Keras, Numpy, OpenCV
Software Tools:	Git, Jupyter, Quartus, Wireshark, AutoCAD, SolidWorks

COURSES UNDERTAKEN

Computer Science : Discrete Structures, Data Structures and Algorithms (+Lab), Data Analysis and Interpretation, Software Systems Lab, Abstractions and Paradigms for Programming (+Lab), Digital Logic and Design (+Lab), Design and Analysis of Algorithms, Logic for Computer science, Computer Networks (+Lab), Computer Architecture (+Lab), Database and Information Systems (+Lab), Operating Systems (+Lab), Automata Theory, Implementation of Programming Languages (+Lab), Artificial Intelligence (+Lab), Fundamentals of Digital Image Processing, Foundations of Intelligent and Learning Agents, Automatic Speech Recognition, Computer Graphics[†].

Miscellaneous : Introduction to Electrical and Electronics Circuits, Linear Algebra, Calculus, Basics of Electricity & Magnetism, Quantum Physics, Engineering Graphics & Drawing ^{† To be completed by Spring 2022}

EXTRACURRICULAR

- Attended **InterIIT Camp** for **Aquatics** and currently a member of **Aquatics Team IITB**
- Competed in **All India IPSC** Swimming Championship (U-19) and won **2nd prize** in an event. (2016)
- Took part in **62nd National School** Swimming Championship (U-19) organised by **SGFI**. (2016)
- Competed in **CBSE West Zone** Swimming Championship (U-16), held in Ajmer. (2015)
- Competed in **62nd State School** Swimming Championship held in Mandsaur, M.P. (2016)
- Helped organize various Lectures at **Techfest**, Asia's largest science and technology festival. (2019-2020)